

CUSTOMER BEHAVIOR ANALYSIS

Data Analytics Internship Project Report

Submitted To	InternSpark (AICTE-Listed Internship Program)
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Candidate ID	IS-2026-14038
Program	B.Tech - Computer Science (Big Data), Presidency University
Internship Domain	Data Analytics
Project Focus	Customer segmentation, purchase patterns, and churn risk analysis
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
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1. Executive Summary

This report analyzes customer transaction behavior for Alfido Tech using the ecommerce customer dataset. The goal is to identify purchase patterns, segment customers using RFM analysis, study churn risk, and recommend actions that can improve customer engagement and retention.

Metric	Value
Total records analyzed	250,000
Unique customers	49,661
Analysis period	2020-01-01 to 2023-09-13
Total revenue	INR 681,346,299
Overall churn rate	19.99%

The analysis shows that revenue is almost evenly distributed across four major product categories, with Home generating the highest revenue. Monthly revenue is stronger from January to August and falls noticeably from September to December. RFM segmentation separates customers into actionable groups such as high value customers, new customers, churn risk customers, and at risk loyal customers.

2. Project Objective

The main objective of this project is to analyze customer transactions and behavior to identify customer segments, purchasing patterns, and churn risks. The final outcome is a set of actionable business recommendations that Alfido Tech can use to improve engagement and retention.

3. Dataset Overview

The dataset contains transaction-level e-commerce records with customer demographics, purchase information, product category, payment method, return status, and churn status. The columns used in the analysis include Customer ID, Purchase Date, Product Category, Product Price, Quantity, Total Purchase Amount, Payment Method, Customer Age, Returns, Gender, and Churn.

Column	Purpose in analysis
Customer ID	Used to group purchases and create customer-level RFM metrics
Purchase Date	Used to calculate recency and monthly purchase trends
Product Category	Used to study category-level revenue and return rates
Total Purchase Amount	Used to calculate revenue and monetary value
Payment Method	Used to understand preferred payment modes
Returns	Used to identify return behavior by category
Churn	Used to evaluate retention and customer risk

4. Data Cleaning and Feature Engineering

Data cleaning ensures the analysis is accurate and consistent. The Purchase Date column was converted into datetime format, missing values in Returns were filled with 0, duplicate rows were removed, and the duplicate Age column was dropped because Customer Age already represented the age information.

Step	Action Taken	Reason
Date conversion	Converted Purchase Date to datetime	Required for time-based analysis and recency calculation
Missing values	Filled missing Returns values with 0	Prevents missing data from affecting return analysis
Duplicates	Removed duplicate rows	Avoids repeated transactions influencing results
Column cleanup	Dropped duplicate Age column	Avoids redundancy and confusion

Feature engineering created additional time-based fields: Year, Month, Day, and Hour. These features make it possible to analyze purchase trends over time and understand when customers are more active.

Feature	Description
Year	Extracted from purchase date to support yearly analysis
Month	Used for monthly revenue and retention trends
Day	Used to examine weekday behavior if required
Hour	Used to examine purchase timing if required

5. Purchase Pattern Analysis

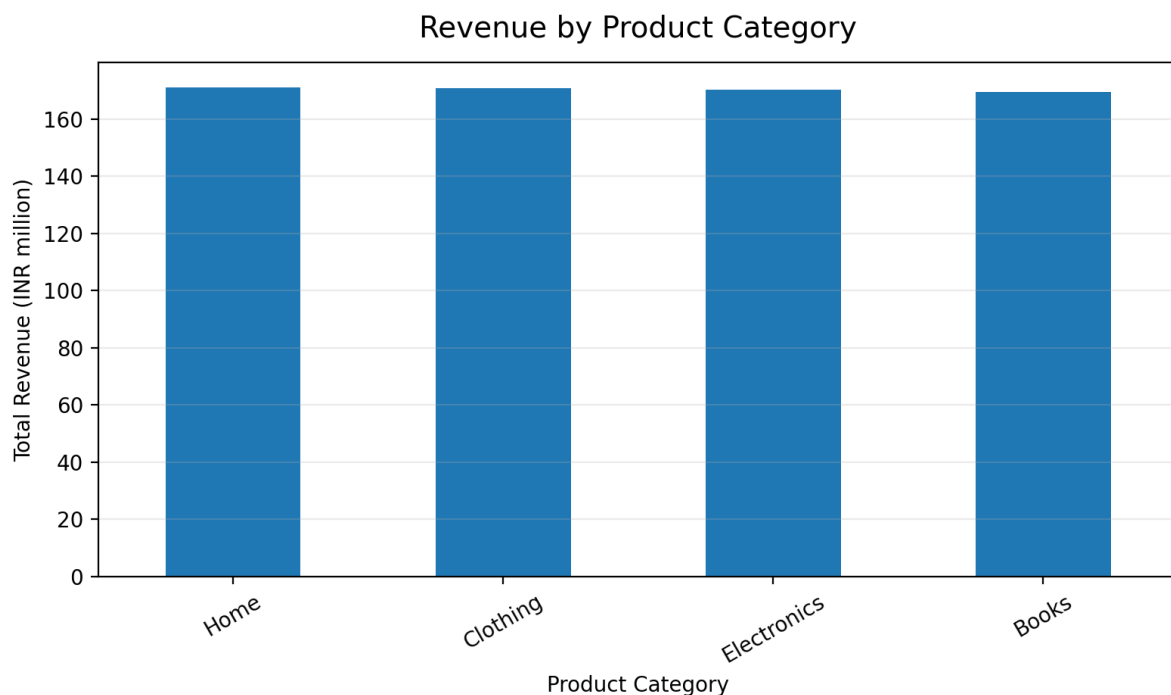


Figure 1. Revenue by Product Category. Home recorded the highest revenue, followed closely by Clothing, Electronics, and Books.

Revenue is distributed very closely across the four product categories. This suggests that Alfido Tech has a balanced product portfolio, but Home and Clothing should still receive slightly higher priority in campaigns because they generate the highest total revenue.

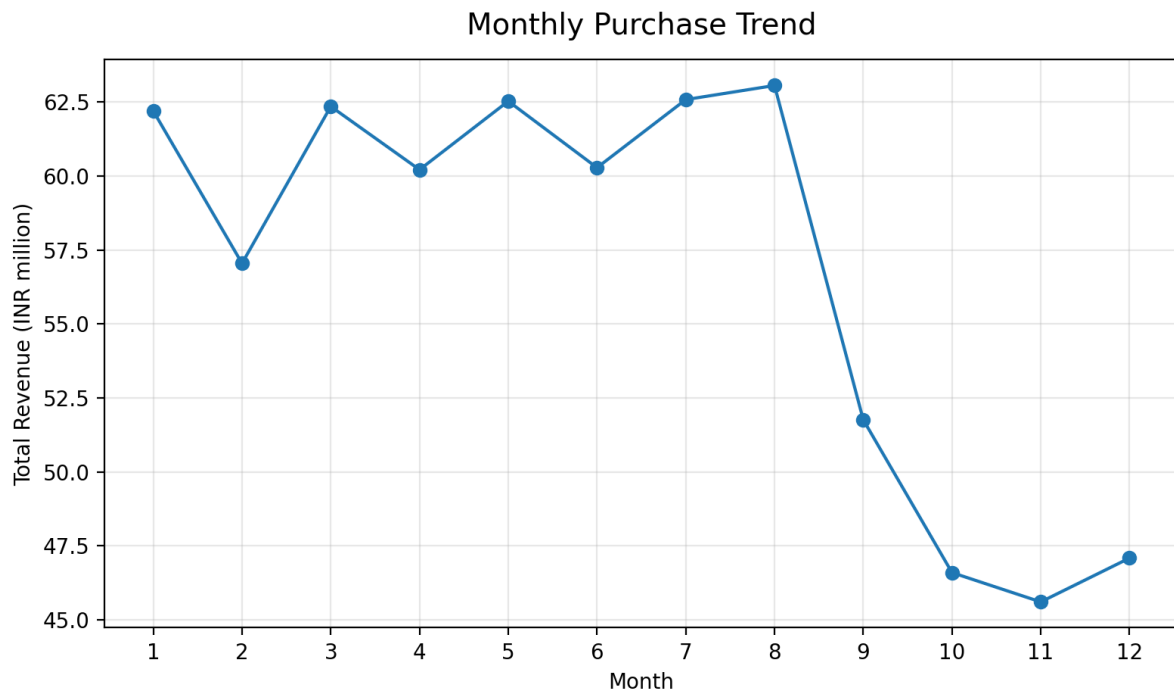


Figure 2. Monthly Purchase Trend. Revenue is stronger from January to August and drops from September onwards.

The monthly trend shows stronger purchase activity during the first eight months of the year, with a sharp fall beginning in September. This gives Alfido Tech an opportunity to plan seasonal campaigns and engagement strategies before the decline begins.

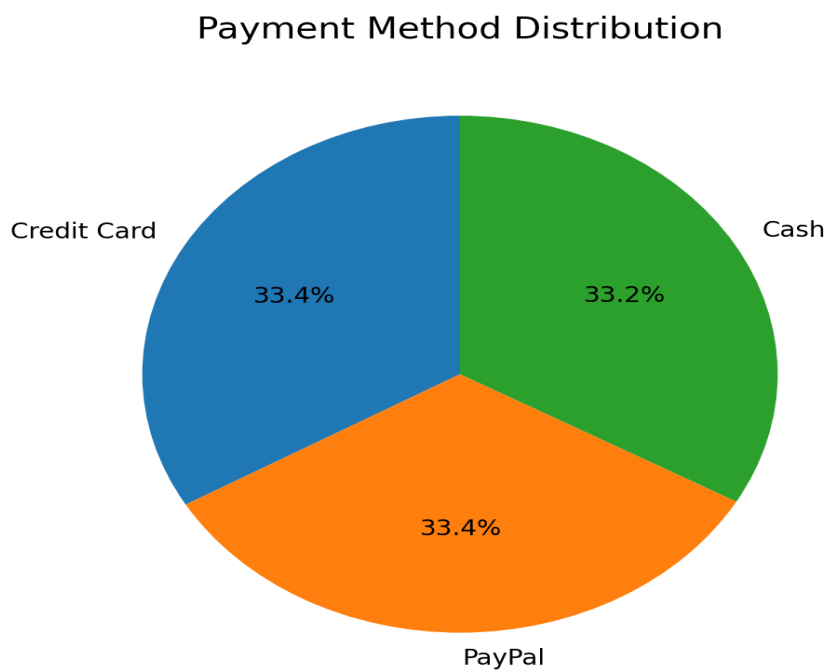


Figure 3. Payment Method Distribution. Credit Card, PayPal, and Cash are used almost equally by customers.

The payment method split is balanced, meaning customers are comfortable with all available payment options. Alfido Tech should continue supporting multiple payment methods and can test payment-specific offers to improve conversion.

6. Customer Segmentation using RFM

RFM segmentation groups customers based on Recency, Frequency, and Monetary value. Recency measures how recently a customer purchased, Frequency measures how often they purchased, and Monetary value measures how much they spent. This technique helps identify customers who deserve different engagement strategies.

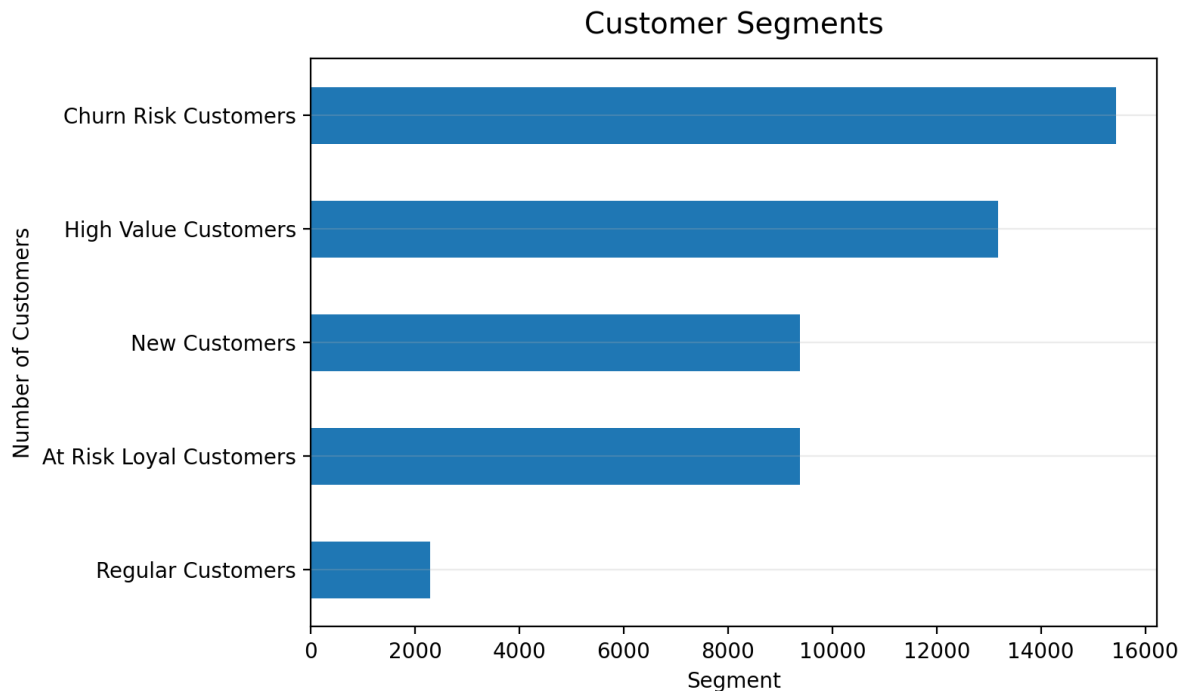


Figure 4. Customer Segments based on RFM analysis. Churn Risk Customers form the largest segment, followed by High Value Customers.

Segment	Count	Avg Recency	Avg Frequency	Avg Monetary	Churn Rate
Churn Risk Customers	15,446	499.44	3.13	INR 8,537	20.00%
High Value Customers	13,178	79.30	7.20	INR 20,297	21.00%
New Customers	9,385	88.09	3.61	INR 9,850	20.00%
At Risk Loyal Customers	9,371	346.95	6.44	INR 17,572	20.00%
Regular Customers	2,281	83.45	5.55	INR 10,916	20.00%

The segment profile shows that High Value Customers purchase most frequently and spend the most, while Churn Risk Customers have the highest average recency and lower purchase frequency. This makes the churn-risk group an important target for reactivation campaigns.

7. Retention and Churn Analysis

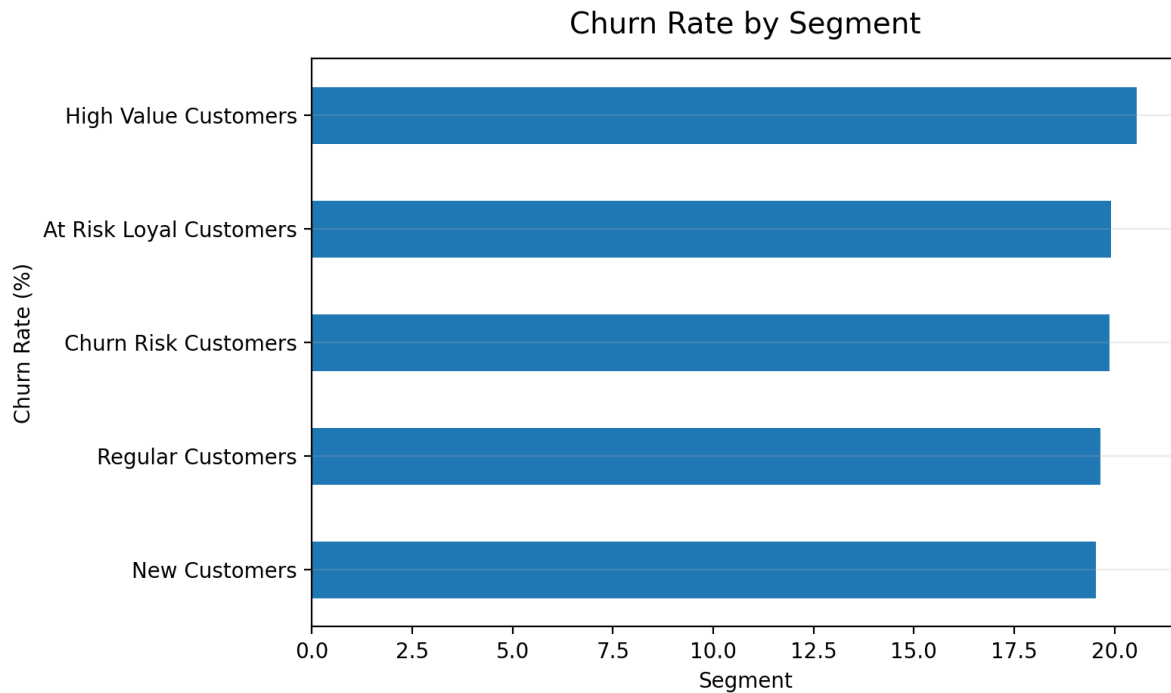


Figure 5. Churn Rate by Customer Segment. Churn rates remain close across segments, but high value customers show a slightly higher churn rate.

The churn rate is close to 20% across most segments. Even though the variation is not very large, losing high value customers can have a greater revenue impact, so retention strategies should prioritize both high value and churn-risk groups.

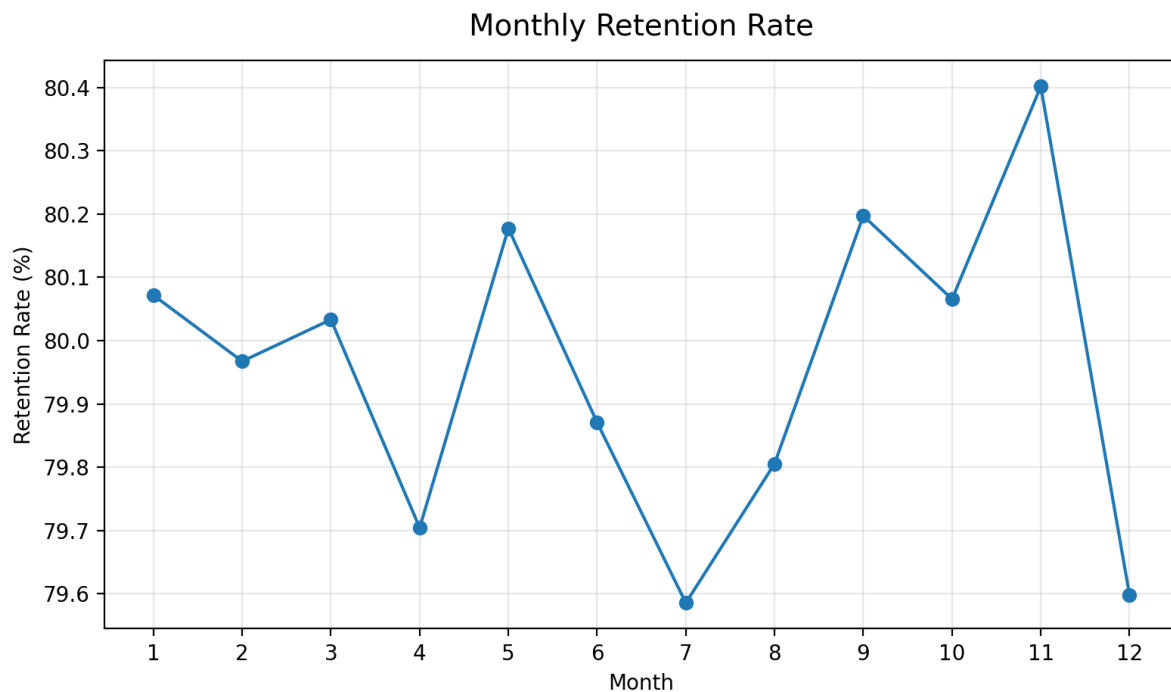


Figure 6. Monthly Retention Rate. Retention stays close to 80% across months.

Monthly retention remains relatively stable, but even small changes in retention matter because the dataset contains a large customer base. Regular engagement through offers, reminders, and personalized campaigns can help improve long-term retention.

8. Return Rate Analysis

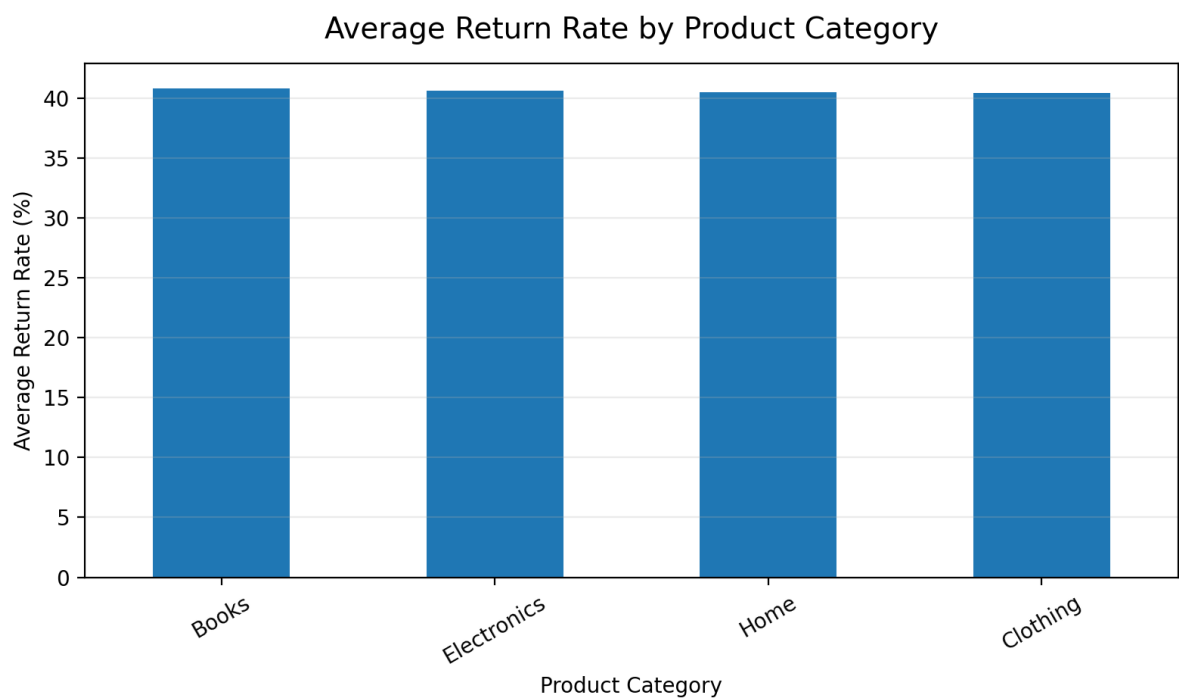


Figure 7. Average Return Rate by Product Category. Return rates are close across product categories, with Books slightly higher.

Return rates are similar across categories, indicating that returns are not limited to one product line. Alfido Tech should still review the product descriptions, images, delivery experience, and quality feedback for categories with higher return rates.

9. Key Findings

- The dataset contains 250,000 transaction records from 49,661 unique customers.
- Total revenue generated in the dataset is INR 681,346,299.
- Home is the highest revenue category, while all four categories contribute almost evenly to revenue.
- Monthly revenue is stronger between January and August and decreases from September onward.
- Payment method usage is balanced across Credit Card, PayPal, and Cash.
- Churn Risk Customers are the largest RFM segment and need immediate re-engagement.
- High Value Customers generate strong spending and should be protected through loyalty programs.

10. Recommendations for Alfido Tech

Recommendation	Actionable Business Use
1. Launch win-back campaigns for churn-risk customers	Send personalized discounts, reminders, and limited-time offers to customers who have not purchased in 30 days.
2. Create a loyalty program for high-value customers	Offer exclusive coupons, early access to sales, or reward points to customers with high frequency and high spend.
3. Strengthen category-based marketing	Prioritize campaigns for Home and Clothing while also cross-selling Books and Electronics to maintain a balanced portfolio.
4. Plan seasonal campaigns before revenue drops	Since revenue falls after August, launch festive or seasonal offers in August and September to maintain momentum.
5. Reduce returns through better product information	Improve descriptions, images, size guides, and customer reviews for categories with higher return rates.

11. Conclusion

This project successfully performed data cleaning, feature engineering, customer segmentation, purchase pattern visualization, retention analysis, and business recommendation development. RFM analysis helped identify meaningful customer groups, while purchase and churn trends revealed opportunities for improved engagement. Alfido Tech can use these insights to target churn-risk customers, reward high-value customers, improve product communication, and plan timely marketing campaigns.

Appendix: Requirement Mapping

Internship Requirement	Where it is addressed in the report
Perform data cleaning and feature engineering	Section 4
Segment customers and profile each segment	Section 6
Visualize purchase patterns	Section 5
Visualize retention trends	Section 7
Provide 5 actionable recommendations	Section 10

Prepared for InternSpark Data Analytics Internship - Task 1.